

Mini Jev: An Inspectable Local Interface for Typed Decisions from Frozen Language Models

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Abstract

Many language-model applications need a small typed decision rather than free-form text. Mini Jev is an open-source local interface that maps a state and explicit criteria to a categorical choice, a true-candidate probability, or an expected ordinal stage. It uses candidate next-token logits from a frozen language model, with no trained decision head. A browser demonstration exposes typed values, candidate distributions, probability semantics, and request/response JSON. A 4,050-request matched comparison finds identical direct-readout and native one-token outputs in all 1,350 pairs, without an established latency advantage. Two subsequent studies across three checkpoints use 7,600 and 14,400 requests to examine presentation sensitivity and probability averaging. Reduced sensitivity can coexist with incorrect or nearly constant predictions, while averaging costs multiple model calls. The contribution is a runnable demonstration and auditable system evaluation, with explicit distinctions among type validity, candidate concentration, task quality, and computation cost.

1 Introduction

A workflow may need a route, an urgency score, or a stage on an explicit rubric. Returning a paragraph and then recovering that value creates an interface problem: the application must validate a result whose useful content is small. A typed interface makes the contract explicit, but a valid type alone says nothing about whether the decision is right. Similarly, a model can assign almost all candidate probability to an incorrect answer.

Mini Jev demonstrates this distinction through a local browser interface and an HTTP/Python API. Users edit a shared state and three questions, run the model, and inspect the resulting values alongside candidate distributions. The interface is inspired by TypeSafe’s Jev (Almeida, 2026; Type-

Safe, 2026b); this independent implementation makes no claim to reproduce that system’s architecture, training, calibration, compatibility, or performance.

The intended users are developers and researchers inspecting small typed decisions from local frozen models before integrating them into a workflow. The mechanism is conventional: a frozen model scores explicit single-token candidates, a restricted softmax produces a distribution, and ordinary code constructs the typed response. Verbalizers, logit-based candidate selection, and candidate-scoring APIs are established (Schick and Schütze, 2021; Ma et al., 2025; LMQL contributors, 2026). Our contribution is a runnable interface for inspecting typed readouts and their probability semantics, together with an auditable evaluation of quality, presentation sensitivity, and computation cost. We do not claim a new scoring method or the first model-inspection interface.

Schema validity, regression checks, external-task quality, and paired runtime measurements answer different questions. Our matched experiment, for example, finds no material latency advantage over native one-token selection. Source, installation instructions, data documentation, and evaluation artifacts: github.com/UpHash-Network/mini-jev. Demonstration video: [approximately 145-second captioned live recording](#).

2 System and Typed Contract

2.1 From a state to typed values

A request contains a shared state and named questions. Each question supplies a type, natural-language instructions, and candidate criteria. The service returns a semantic label, a candidate distribution, and a type-specific value:

- **Choice** returns the key of the most likely candidate. Keys are canonically sorted before prompting.

- **Noul** exposes the probability of the true candidate in a two-candidate false/true decision.
- **Score** returns the expected zero-based stage of an ordered rubric; the most likely stage is also available as the label.

The distinction between Score’s expectation and its hard label matters. An expectation of 1.6 on stages 0–3 is a probability-weighted position, not an additional model-generated stage or a calibrated utility. Its interpretation assumes equally spaced stage indices; users must decide whether that fits their rubric.

Let z_i denote the next-token logit for candidate $i \in \{0, \dots, K-1\}$, with K candidates and temperature $T > 0$. The service computes

$$p_i = \frac{\exp(z_i/T)}{\sum_{j=0}^{K-1} \exp(z_j/T)},$$

$$s = \sum_{i=0}^{K-1} i p_i.$$
(1)

These are probabilities conditional on the selected candidate set. They are neither probabilities over all valid textual answers nor guarantees that the criteria are exhaustive. Changing the criteria changes the question as well as the normalization. The optional scalar temperature preserves the maximum-logit label, but can change Noul and Score values.

The interface also displays entropy concentration,

$$c = 1 - H(p)/\log K,$$

$$H(p) = - \sum_i p_i \log p_i.$$
(2)

A concentrated distribution has c near one. The display identifies this quantity as concentration rather than estimated accuracy. The response includes probability-semantics and calibration metadata, so downstream applications can retain the same distinction.

2.2 Local execution

The released inference configuration uses Qwen3.6-35B-A3B in Q4_K_M GGUF form (Qwen Team, 2026; ggml-org, 2026), through a source-built llama.cpp helper (ggml-org and contributors, 2026). The model and its output projection remain frozen; there is no trained decision head. The helper performs the full vocabulary projection and then gathers candidate logits. It

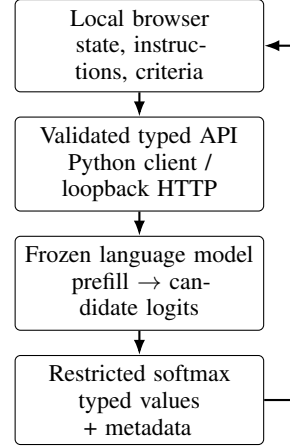


Figure 1: The live demonstration uses the released typed API. It displays model-derived candidate distributions and typed values; no generated explanation or simulated prediction is inserted.

does not implement a selected-row projection optimization.

The prompt repeats the task twice, following the use of repetition for non-reasoning inference (Leviathan et al., 2025). Choice and Noul use letter labels; Score with at most ten stages uses digits after a prefilled answer-object prefix. Larger Score rubrics use letters. The implementation checks that every candidate is a distinct single token at the actual prefix boundary. The API accepts 2–26 candidates; quality evaluation covers 2–8, while larger candidate sets have functional coverage only.

A multi-question request executes its questions sequentially. Model state is cleared between questions: they share the supplied state text, but not a cached prefill. Complete inputs exceeding 2,048 tokens are rejected rather than silently truncated. Schema validation and deterministic response construction prevent free-form output from crossing the API boundary; these software properties do not prevent semantic errors.

2.3 Relationship to one-token decoding

Restricted softmax readout has the same mathematical candidate distribution as a sampler that masks all non-candidates, provided the model prefix, logits, candidate IDs, temperature, and other processors match. Greedy labels then agree unless tie-breaking differs; stochastic sampling agrees in distribution, not necessarily in realized samples. This is an equivalence under specified conditions, not a new inference theorem or learning method.

In particular, selecting a first token after pre-

fill need not require another model forward pass. Avoiding a generated token therefore does not itself establish a speedup. The matched study in Section 4 tests this point using an actual native one-token sampler, rather than comparing direct readout only with a long textual answer.

3 Demonstration and Access

The browser runs locally and connects through a small Python bridge to the existing model API. It requires no cloud inference service. A user starts with a shared state and three editable questions: routing through Choice, urgency through Noul, and priority through Score. The question instructions and criteria are visible, making the decision contract available for inspection before inference.

The user first defines the state and candidate meanings, then selects *Run decisions*. The displayed probabilities let the user compare a hard label with its distribution; Score’s expected and most likely stages can differ. Editing a state or criterion and rerunning explores behavior, but changes the model input and is not a controlled accuracy experiment.

The page also displays entropy concentration, elapsed time, and copyable request/response JSON. The bars show relative candidate mass, the JSON exposes the contract, and elapsed time describes the live request; none is a reference answer. Editing marks prior results stale. There is no replay mode: an unavailable model API produces an error, not a canned prediction.

Installation begins with the repository’s native build script and pinned model downloader. Users build the helper with `build_native.sh`, download the model with `fetch_native_model.py`, and start the API with `./run.sh -without-calibration`. Starting `python3 -m demo.server -port 8766` then opens access at <http://127.0.0.1:8766/>. The repository supplies complete commands and a Python client example. Fresh builds start at $T = 1$; historical calibration artifacts are bound to their recorded runtime and should not silently transfer to a different build.

The tested environment is an Apple M5 Pro with 64 GB unified memory and macOS 26.4, using Metal and Accelerate. The model file is approximately 20.4 GB, in addition to runtime memory. This is a local workstation demonstration, not a claim of lightweight deployment or a demon-

Evidence	Size	Interpretation
Local regression	2,400 items	Self-authored Japanese; 800 per type
Expanded external	600 items	JCoLA, JSTS, JCQA; 200 each
Matched run-time	4,050 quests	re- 150 local $\times$ 5 repetitions, plus 600 external; 3 modes
Presentation study	7,600 quests	re- Same 600 external items; 3 checkpoints
Cyclic averaging	14,400 requests	600 additional external items; 3 checkpoints

Table 1: Evaluation accounting. The last two studies use 1,200 unique source items in total. Measured counts exclude synthetic checks and warm-ups. JCQA abbreviates JCommonsenseQA.

strated minimum-memory requirement. Source is distributed; model weights and compiled native binaries are not bundled.

4 Evidence Behind the Demonstration

4.1 Evaluation panels and provenance

Table 1 separates the evidence sources. Repeated requests are not additional quality examples, and the external tasks do not share one accuracy metric.

The historical local suite contains 2,220 generated items from 45 template families and 180 individually AI-authored items; its legacy manual field does not mean human annotation. Authoring used earlier development errors, so it is regression material rather than a blind external test. All 2,400 outputs were valid, but 162 top labels were wrong; structural validity does not establish correctness. Historical temperature fitting also did not improve all calibration and score-error metrics.

The expanded sample has 200 public-development examples each from JCoLA (Someya et al., 2024), JSTS, and JCommonsenseQA (Kurihara et al., 2022). Hash selection and rubrics were frozen before inference, without independent preregistration. Six JSTS rows share text or image provenance with an earlier 300-item JNLI pilot. Public development data may have appeared in model training; these are not independent hidden-test evaluations.

4.2 Matched runtime and parity

All three conditions used one resident model and a common research HTTP response containing a type and semantic label. Direct readout and native one-token greedy selection used identical com-

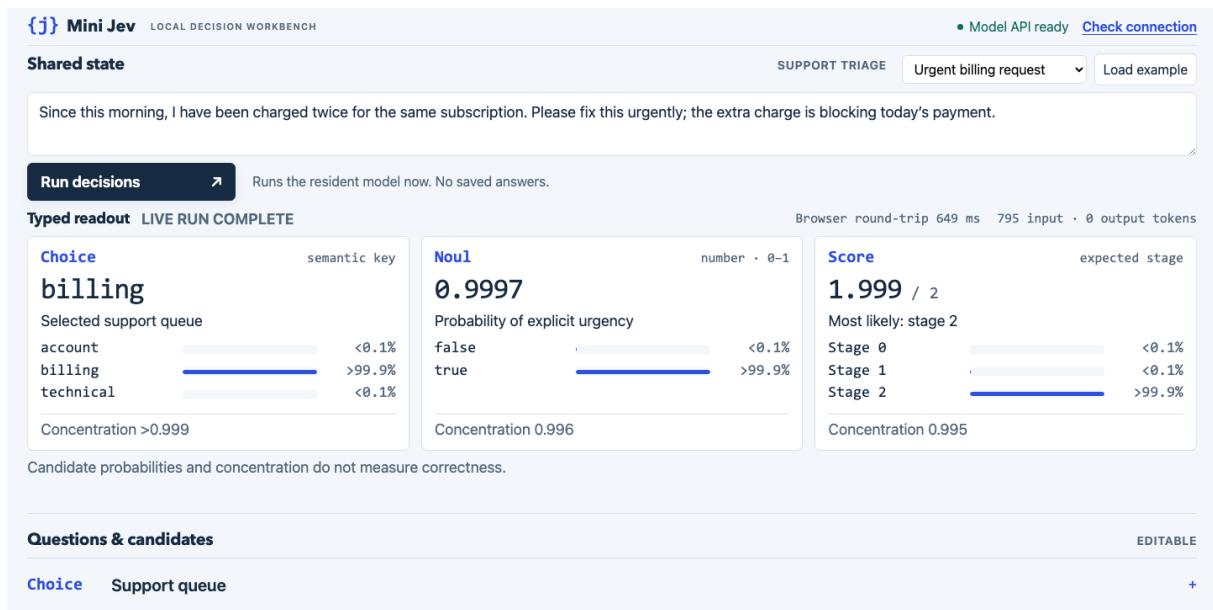


Figure 2: Live browser demonstration in presentation view: an editable state, all three typed outputs, and their candidate distributions. The question editors continue below the displayed results. This English example illustrates the interface; it is not an evaluation item or evidence of English task quality.

plete prompts and candidates. The JSON condition actually generated a grammar-constrained object, with a serialization-specific prompt. Its timing comparison therefore includes prompt and pre-fill differences as well as multi-token generation; it does not isolate sampler overhead.

The schedule contained $(150 \times 5 + 600) \times 3 = 4,050$ measured requests, plus 21 synthetic warm-ups. All completed with valid responses and no recorded HTTP failures. Mode order was randomized within each item/repetition. Timing covered serialization through loopback HTTP parsing, including native audit transfer but excluding trace-file writes. These are research-harness timings, not browser rendering times.

Direct and one-token conditions matched prompts, token sequences, candidate IDs, logits, and labels in all 1,350 paired observations. Maximum observed candidate-logit difference was zero; there were no maximum-logit ties. The primary latency statistic is the median across 150 local items of the within-item difference between five-repetition mean latencies. It is not a difference between marginal medians.

Table 2 gives 10,000-replicate whole-family/tag percentile intervals. They describe sensitivity to the observed cluster composition, conditional on this model, machine, selected items, and session. They do not estimate between-machine or independent-session variability. The one-token

Comparator – direct	Difference	95% interval
One token	+0.0898	[−0.7483, 0.8449]
JSON	+160.2958	[138.1189, 177.0587]

Table 2: Primary paired HTTP latency differences in milliseconds. Intervals resample observed local family/tag clusters; JSON uses a different serialization prompt.

interval crosses zero: this run provides no evidence of a material direct-readout latency advantage. JSON was slower under its different serialization requirements, with task-dependent quality changes.

4.3 External task-specific results

Direct readout and one-token selection have identical external labels. JCoLA reached 86.0% accuracy and MCC 0.5708; JSON reached 82.0% and MCC 0.5160. The selected sample contains 158 acceptable sentences, so an always-acceptable rule already achieves 79.0% accuracy and has undefined MCC. Reporting only accuracy would obscure this imbalance.

JCommonsenseQA accuracy was 95.0% for direct/one-token and 95.5% for JSON, a difference of one example. JSTS retains its continuous reference scores in $[0, 5]$: direct expected-stage MAE at $T = 1$ was 0.5065. JSON has no comparable probability vector, so we do not assign it an expected-

stage metric. The separate hard-stage comparison gives MAE 0.5480 for direct/one-token and 0.5650 for JSON. We do not round the gold scores into invented classes or pool these tasks into one accuracy.

4.4 Presentation sensitivity and averaging

Two subsequent studies add Qwen2.5-0.5B-Instruct and Qwen2.5-1.5B-Instruct to the native 35B checkpoint. The first uses 7,600 requests on the same 600 external items. All 1,800 exact-input repeat pairs have identical logits and probabilities, yet display reversal changes native JCQA labels on 11/200 items. Conversely, the 1.5B checkpoint has zero JCoLA flips while always predicting acceptable: stability alone is insufficient.

The second freezes 600 additional public-development items before its 14,400 requests. It excludes prior IDs and literal text reuse, and selects JSTS pairs from previously unused sentence/image components. However, 194/200 JCoLA items share bibliographic sources with earlier evaluations. Both studies complete without inference errors; each excludes 57 synthetic checks/warm-ups.

We average semantic candidate probabilities at $T = 1$ across cyclic rotations of complete option lines, keeping output-token–meaning bindings fixed. Fixed-binding transformations and cyclic aggregation have precedent (Wei et al., 2024; Zheng et al., 2024); this is not a new algorithm or a reproduction of PriDe. Table 3 compares canonical-single and forward-cyclic accuracy, alongside single/reversed and forward/reverse-cyclic label disagreement. Lower disagreement does not consistently improve accuracy. The 0.5B cyclic JCoLA output is always acceptable (78.5% accuracy, undefined MCC). On JSTS, 0.5B and 1.5B cyclic hard labels occupy one stage on 199/200 and 200/200 items, respectively. These marginals expose collapse hidden by type validity or stability.

One cyclic answer costs 2, 5, or 6 model calls for JCoLA, JCQA, or JSTS; the single baseline reuses a pool member. Binary forward/reverse pools contain the same two calls, so their equality is structural. There is no equal-call repeat control in this averaging study. The small checkpoints use Transformers/MPS float32; the native model uses quantized Metal inference. Architecture, precision, and backend differ, precluding a model-size or cross-backend speed conclusion. These

Checkpoint	Accuracy (%)	Flip rate (%)
Qwen2.5 0.5B	39.5 → 36.0	65.5 → 30.5
Qwen2.5 1.5B	80.0 → 83.5	26.0 → 12.5
Qwen3.6 35B-A3B	98.5 → 98.0	4.5 → 1.5

Table 3: JCQA on the additional 200-item panel: single → cyclic point estimates. Flip rate measures orientation disagreement, not error rate. Native 35B uses Q4_K_M. These are not equal-compute comparisons.

are descriptive system checks; fuller distributions, grouped intervals, and call accounting accompany the artifacts.

4.5 Reproduction and framework integration

The versioned source/evidence package at github.com/UpHash-Network/mini-jev records the three studies’ 26,050 requests, protocols, selections, hashes, analyzers, and AI-agent audits. Clean-directory CPU reanalysis reproduced all 56 derived files byte-for-byte. A fresh native build served all three types using a cached, hash-verified model on the same Mac. This validates reconstruction with source/model cache reuse, not a second-machine or fresh-download replication.

Separate synthetic diagnostics test integration boundaries. With the same checkpoint, complete prefixes and candidate IDs, unmodified LMQL 0.7.3 scoring through a custom native backend matched all 12 direct-readout labels. Maximum probability difference was 2.63×10^{-5} ; three cases exceeded the prespecified 10^{-5} tolerance. Different native scoring paths therefore did not establish exact numerical parity. ChainForge 0.3.7.4’s registration/dispatch routes, with a custom provider calling the released API, preserved all eight typed-answer pairs across four request pairs (maximum probability difference zero); three invalid inputs were rejected. This checks backend interoperability, not the browser flow editor, human usability, or task quality. These diagnostics are separate from the three evaluation panels. Inference replication still requires upstream data, pinned models, and compatible runtimes; hardware/build equivalence is not guaranteed.

5 Related Work and Scope

Scoring and structured outputs. Verbalizers (Schick and Schütze, 2021), decoding-free candidate selection (Ma et al., 2025), Qwen3 yes/no reranking (Zhang et al., 2025; Qwen Team, 2025), and G-Eval’s probability-weighted scores (Liu

et al., 2023) precede our readouts. LMQL exposes continuation scoring, normalization, and maximum selection (LMQL contributors, 2026); general multi-token continuation scoring need not equal our single-token readout. Outlines and Guidance provide constrained choices and structured generation (dottxt and Outlines contributors, 2026; Guidance contributors, 2026), as does grammar-constrained decoding research (Geng et al., 2023). TypeSafe’s official adapter already maps general LLM outputs to a Jev-style contract (TypeSafe, 2026a). Its generated probability values should not be conflated with token-derived candidate probabilities. Our JSON experiment is a narrow serialization comparison, not a benchmark of these frameworks.

Inspection interfaces. ChainForge supports visual prompt experimentation (Arawjo et al., 2024); LLM Comparator supports interactive side-by-side response evaluation (Kahng et al., 2025). Thus visualization and local inspection alone are not new. Mini Jev focuses on a shared Choice/Noul/Score contract with visible candidate probabilities, expected stages, concentration, and request/response JSON. Its live interface does not implement the research harness’s cyclic averaging or multi-run comparison. We have not established a usability advantage over these tools.

Sensitivity and calibration. Order/token bias and fixed-binding changes are established (Wei et al., 2024; Zheng et al., 2024). Hanna and Feng (2026), a recent preprint, also report that reduced order sensitivity need not improve accuracy; we do not claim that distinction as a new general finding. Our contribution is its concrete examination alongside typed outputs and measured call costs. Temperature scaling (Guo et al., 2017) differs from contextual label-bias correction (Zhao et al., 2021). Canonical key sorting ensures software equivalence under dictionary insertion changes, not model invariance to option meaning or placement. The frozen 35B configuration establishes neither training gains nor superiority over other typed-output systems.

Limitations

Presentation studies cover three checkpoints on one Apple Silicon device; the matched parity/latency comparison covers only the native quantized model in one session. Different ar-

chitectures, precisions, and backends prevent attribution to parameter count. Quality data are Japanese; an English interface example does not establish English task quality. Public subsets have unknown training contamination and remaining source dependence. Local data are AI-authored and development-related, without independent human reference validation. There is no user study establishing usability or downstream decision quality.

Candidate probabilities depend on prompts, labels, candidate coverage, and temperature. Concentrated wrong answers occur; neither type validity nor entropy concentration should trigger consequential actions without application-specific validation. Score assumes equally spaced stages and Noul is a relative candidate probability, not a verified frequency of truth. The model is large, questions run sequentially, and other operating systems or lower-memory configurations have not been validated. Live-demo elapsed time is not directly comparable with the research harness’s controlled interval.

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Ethics and Availability

The demonstration is intended for inspecting model behavior and prototyping typed interfaces. It provides no evidence for high-stakes deployment. Live inference is local; users still remain responsible for the text they submit and any downstream use. The released code and self-authored local data use MIT terms. External-dataset-derived selection metadata, rubrics, protocols, and results retain the applicable CC BY-SA 4.0 attribution and share-alike terms, separately from generic code. Upstream dataset text, model weights, and compiled binaries are not bundled; the repository’s data cards and notices specify acquisition and attribution.

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